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School of Ai & Future Technologies

Summer Internship Weekly Report

Student Details

Student Name: Himanshu Pathak	**Roll No.: UAI05BT12417**
Year/ Semester: 2nd Year / 4th Semester	**Email Id: himanshu.pathak@universalai.in**
Faculty Mentor : Prof. Rajesh Bhise	

Company Details:

Company Name: Ai Cell, Universal Ai University	**Department Name: School of Ai & Future Technologies**
Company Manger Name: Prof. Rajesh Bhise / Dr. Shivaji Pawar	**Official Mail Id:**
Contact No:	

Project Title: UAi PsychePortal & AI Productivity Analyzer Suite

Week No.: First Week

Reporting Period: 4-May-2026 to 9-May-2026

1. Weekly Task

S. No	Task Description	Status
1	Requirement analysis, architectural design, and repository layout setup	Completed
2	Defined SQLAlchemy database schema (User, StudentProfile, Assessment)	Completed
3	Implemented Flask factory pattern and dynamic configuration handling	Completed

2. Technologies/Tools Used

- **Python 3.13, Flask 3.0.0**
- **Flask-SQLAlchemy, SQLite**
- **Git, VS Code**

3. Learning Outcomes

- Standardized database normalization techniques for relational models storing psychometric profiles.
- Configured Flask application factory patterns to isolate development and production environments.

4. Challenges Faced & Solutions

Challenges: Avoid cyclic imports between models, database initializers, and authentication blueprints.

Solutions: Implemented the Application Factory pattern (create_app) in app/__init__.py, decoupling model declarations from route handlers.

5. Plan for Next Week

S. No	Task to be done	Estimated Time
1	Implement DASS-21, BFI-44, EQ, and SQ mathematical scoring functions	6 Hours
2	Create automated risk-flagging logic for severe student scores	4 Hours
3	Develop seed data scripts (seed_questions.py, seed_demo_data.py)	4 Hours

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Project Title: UAi PsychePortal & AI Productivity Analyzer Suite

Week No.: Second Week

Reporting Period: 11-May-2026 to 16-May-2026

1. Weekly Task

S. No	Task Description	Status
1	Implemented scoring algorithms for DASS-21, BFI-44, EQ, and SQ in app/scoring.py	Completed
2	Engineered automated emergency risk flagging for Moderate/Severe subscale categories	Completed
3	Created database population scripts for 100+ demo profiles and question banks	Completed

2. Technologies/Tools Used

- **Python 3.13, SQLAlchemy ORM**
- **Flask-Bcrypt, Flask-Login**
- **SQLite database browser**

3. Learning Outcomes

- Mastered reverse-score transformations (6 - R) and subscale matrix indexing for Big Five inventory metrics.
- Developed automated event triggers in SQLAlchemy to flag high-risk entries upon submission.

4. Challenges Faced & Solutions

Challenges: Handling missing item selections without crashing subscale calculations.

Solutions: Defaulted missing responses to neutral values (3) for BFI-44 and 0 for DASS-21 inside safe map retrievals (resp_map.get(i, default)).

5. Plan for Next Week

S. No	Task to be done	Estimated Time
1	Construct programmatic PDF report builder using ReportLab	6 Hours
2	Implement Excel data export using Pandas and OpenPyXL	4 Hours
3	Build Student Survey Web Portal (student-survey-portal/app.py)	5 Hours

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Project Title: UAi PsychePortal & AI Productivity Analyzer Suite

Week No.: Third Week

Reporting Period: 18-May-2026 to 23-May-2026

1. Weekly Task

S. No	Task Description	Status
1	Developed PDF export generation using ReportLab (SimpleDocTemplate, TableStyle)	Completed
2	Built Excel export features for counsellor metrics using Pandas and OpenPyXL	Completed
3	Deployed standalone Flask Student Survey Portal running on port 5001	Completed

2. Technologies/Tools Used

- **ReportLab 4.1.0, Pandas 2.1.4**
- **OpenPyXL 3.1.2**
- **Flask Jinja2 Templating**

3. Learning Outcomes

- Standardized programmatic document formatting using ReportLab flowables, tables, and page breaks.
- Integrated dual-Flask microservice instances sharing a common SQLite relational database.

4. Challenges Faced & Solutions

Challenges: PDF layout text overflow when rendering student assessment details.

Solutions: Encapsulated tabular cell text within Paragraph flowables with automated auto-wrapping text styles.

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5. Plan for Next Week

S. No	Task to be done	Estimated Time
1	Develop modern React SPA Counsellor Dashboard using Vite and TailwindCSS	8 Hours
2	Build 11 dashboard view modules (Risk Alerts, Directory, Deployer)	5 Hours
3	Set up Chart.js radar and trend analytics visualizations	4 Hours

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Project Title: UAi PsychePortal & AI Productivity Analyzer Suite

Week No.: Fourth Week

Reporting Period: 25-May-2026 to 30-May-2026

1. Weekly Task

S. No	Task Description	Status
1	Built Vite + React single-page application inside uai-counsellor-dashboard/	Completed
2	Designed 11 dynamic views featuring TailwindCSS styling and Lucide icons	Completed
3	Configured interactive Chart.js visualizations for subscale trend analysis	Completed

2. Technologies/Tools Used

- **React 18, Vite 5**
- **TailwindCSS 3.4, Lucide-React**
- **Chart.js 4.4**

3. Learning Outcomes

- Engineered dynamic dashboard views using React state hooks (useState, useEffect).
- Applied responsive dark-theme design patterns using TailwindCSS utilities.

4. Challenges Faced & Solutions

Challenges: Integrating client-side React routes into Flask server-rendered endpoints.

Solutions: Compiled Vite static assets into Flask's static directory and rendered the root React via Jinja2 templates.

5. Plan for Next Week

S. No	Task to be done	Estimated Time
1	Inject Flask SQLite data directly into React global variables (window.__FLASK_KPIS__)	4 Hours
2	Develop ProductivAI synthetic dataset generator (50,000 samples)	4 Hours
3	Train and serialize Random Forest ML productivity regression model	5 Hours

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Project Title: UAi PsychePortal & AI Productivity Analyzer Suite

Week No.: Fifth Week

Reporting Period: 1-Jun-2026 to 6-Jun-2026

1. Weekly Task

S. No	Task Description	Status
1	Connected React frontend with Flask database using Jinja2 JSON script injection	Completed
2	Built 50,000-sample synthetic lifestyle dataset generator (generate_dataset.py)	Completed
3	Trained Random Forest Regressor and exported model via Joblib (productivity_model.pkl)	Completed

2. Technologies/Tools Used

- **Scikit-Learn 1.3, Joblib**
- **NumPy, Pandas**
- **FastAPI, Uvicorn**

3. Learning Outcomes

- Evaluated regression models using metrics such as R^2, MAE, and MSE.
- Deserialized trained .pkl estimators inside FastAPI endpoints for real-time inference.

4. Challenges Faced & Solutions

Challenges: Model inference latency spikes when serving REST requests under heavy load.

Solutions: Cached the deserialized Joblib model instance in global application memory upon FastAPI startup.

5. Plan for Next Week

S. No	Task to be done	Estimated Time
1	Develop ProductivAI React frontend featuring score dials and Recharts	5 Hours
2	Write integrated multi-process launcher scripts (launch_all.bat / .ps1)	4 Hours
3	Perform end-to-end system testing, bug fixing, and documentation writing	4 Hours

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Contact No:	

Project Title: UAi PsychePortal & AI Productivity Analyzer Suite

Week No.: Sixth Week

Reporting Period: 8-Jun-2026 to 13-Jun-2026

1. Weekly Task

S. No	Task Description	Status
1	Built ProductivAI React analytics dashboard with score dials and dark/light themes	Completed
2	Created unified multi-process orchestrators (launch_all.bat and launch_all.ps1)	Completed
3	Executed full system integration verification across ports 5000, 5001, and 8000	Completed

2. Technologies/Tools Used

- **Windows Batch Scripting, PowerShell**
- **React 18, Recharts 2.10**
- **Framer Motion**

3. Learning Outcomes

- Automated multi-process application deployment using shell scripting techniques.
- Verified cross-origin data flows and API request validation across distinct microservices.

4. Challenges Faced & Solutions

Challenges: Managing multiple terminal windows and port collisions during full-stack integration.

Solutions: Wrote automated PowerShell start routines with explicit port bindings (-port 5000, -port 5001, -port 8000).

5. Plan for Next Week

S. No	Task to be done	Estimated Time
1	Final project deliverable submission and demonstration to faculty mentors	0 Hours

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